ADITYA BHATI

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EDUCATION

The Ohio State University – Main Campus *Expected December 2026*
Undergraduate in Computer Science & Engineering **GPA: 3.43/4.00**

RELEVANT COURSEWORK

Data Structures & Algorithms | Computer Architecture | Operating Systems | Discrete Structures | Statistics & Probability | OOP in C++ | Database Systems | Machine Learning | Cloud Data Management | Computer Vision | Speech & Language Processing

EXPERIENCE

Able Applied Technologies Columbus, OH
Software Engineering Intern *June 2026 – August 2026*

- Built two Python services from scratch that let non-technical staff verify ∼3,200 deployed LCD signage and gateway setups at gas stations without querying the database; joins vendor API data with multi-table SQL to generate on-demand reports in ∼20 seconds.
- Containerized both services (Docker) and deployed them on Amazon ECS behind a single Route 53 domain (Lambda landing page, ALB path-based routing) so staff generate a report by setting parameters in a browser; credentials pulled from Secrets Manager via a custom IAM role.
- Developed a systemd Python daemon using pyserial to poll an LCD sign every 5 minutes, 24/7, logging brightness and temperature alongside local weather data to enable long-term stress-test monitoring that didn't exist before.
- Implemented a React/TypeScript dashboard (Tailwind CSS) for that data with filterable charts, drilldown, and adjustable time granularity, plus CSV re-upload to refresh visualizations with newly collected sign data; deployed on AWS Amplify.

PROJECTS

Damage Detector | Python, PyTorch, YOLOv8, FastAPI, React, TS *September 2026 – October 2026*

- Fine-tuned a YOLOv8s object detection model (PyTorch) on CarDD to detect six types of car damage (0.745 test mAP50); deployed it as a Dockerized FastAPI API on Google Cloud Run with a React/TypeScript front end on Vercel.
- Built a 200-image clean-car eval set (CompCars) as the benchmark couldn't measure false alarms; adding 280 clean-car negatives cut them from 66% to 14% with recall unchanged. Added a live user-reported accuracy counter via Firestore.

FitLog | React, TS, HTML, CSS, Firebase *June 2025 – Present*

- Designing and developing a full-stack CRUD application in React/TypeScript where users publish, like, and comment on posts with outfit images and descriptions, using client-side image compression and a responsive layout.
- Backed by Firebase (Auth, Firestore NoSQL, Storage); enforces unique usernames with atomic batched writes for concurrency safety and field-level security rules, verified by an automated Vitest suite run against the local emulator.

QuantiFiAI – SIAM2I 2025 Quantathon | Python, PyTorch, Sci-Kit Learn, Pandas *March 2025*

- Developed a quantitative trading system in Python employing PyTorch LSTM and attention-based neural networks, a scikit-learn gradient boosting classifier with engineered features, and anomaly detection for market regime prediction, beating buy-and-hold S&P 500 from 2019–2022.
- Designed an ensembled anomaly detection approach (Isolation Forest, DBSCAN, z-score) and optimized volatility-targeting, stop-loss, and trend-filter parameters, securing 2nd place at SIAM2I's 2025 Quantathon at OSU.

Artist Search Website | Python, HTML, CSS, JS, Flask *November 2024 – January 2025*

- Leveraged Spotify's RESTful Web API via Python with OAuth Client Credentials auth, chaining four endpoint calls to acquire an artist's follower count, genres, top songs, and discography.
- Utilized Flask and Jinja2 to present formatted back-end data (follower counts, h:m:s track durations) on an HTML, CSS, and JavaScript page with collapsible accordions.

TECHNICAL SKILLS

Languages: C/C++ | Java | JavaScript | TypeScript | Python | HTML/CSS | SQL
Frameworks & Tools: React | Tailwind | REST APIs | PyTorch | scikit-learn | Firebase/Firestore | MySQL | AWS | GCP | YOLOv8 | Docker | Claude Code